1 Simulation Study

We assess calibration and parameter recovery for the revised Python model using two fixed offline training budgets, 3,000 and 30,000 simulations. Both infer (w, r, v, η) with the priors and observation channels of the variable-radius reference in **??**: inputs contain trajectories and neighbor counts at fixed reference distances, excluding statistics evaluated at the unknown true radius. The revised simulator uses relative-heading steering and additive diffusive heading noise. Each trial contains 49 agents observed for 60 seconds at $\Delta t = 0.1$ seconds.

Both estimators use the same transformer summary network and flow-matching posterior architecture as the extension reference. The smaller training bank is a subset of the larger bank; the runs share 300 independent validation simulations and 300 independent test simulations, with $1,000$ posterior draws per test case. Training uses one initialization seed, batch size 32, and at most 300 epochs, with validation-loss early stopping after 20 epochs without improvement. The selected epochs are 23 and 35 for the smaller and larger budgets, respectively. Each estimator reuses its fixed bank across epochs; the extension reference instead consumes 480,000 fresh online simulations. These runs do not provide an equal-budget comparison of offline and online training.

1.1 Computational Faithfulness: Testing Calibration

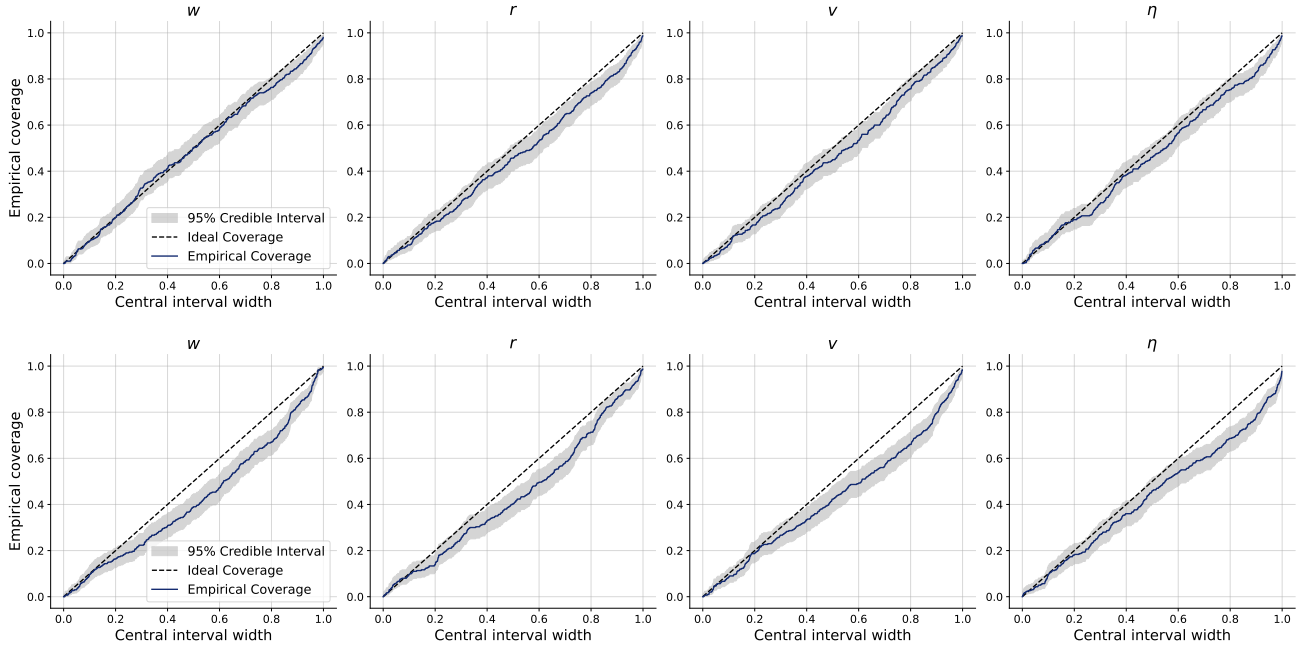


Figure 1: Empirical coverage for the revised offline estimators trained on 3,000 (top) and 30,000 (bottom) simulations.

We evaluate computational faithfulness using empirical coverage of marginal credible intervals. Expected calibration error (ECE) averages the absolute difference between empirical and nominal coverage over the diagnostic's interval levels; lower values indicate closer agreement. Figure 1 shows the coverage curves and Table 1 reports ECE. From the smaller to the larger budget, ECE changes as follows: w: 0.008 to 0.084, r: 0.043 to 0.050, v: 0.038 to 0.076, η: 0.036 to 0.052. For this training seed, calibration error increases for all four parameters. The larger-budget coverage curves generally lie below nominal coverage, indicating overly confident marginal posteriors. The gains in recovery therefore coexist with weaker calibration.

1.2 Model Sensitivity: Testing Parameter Recovery

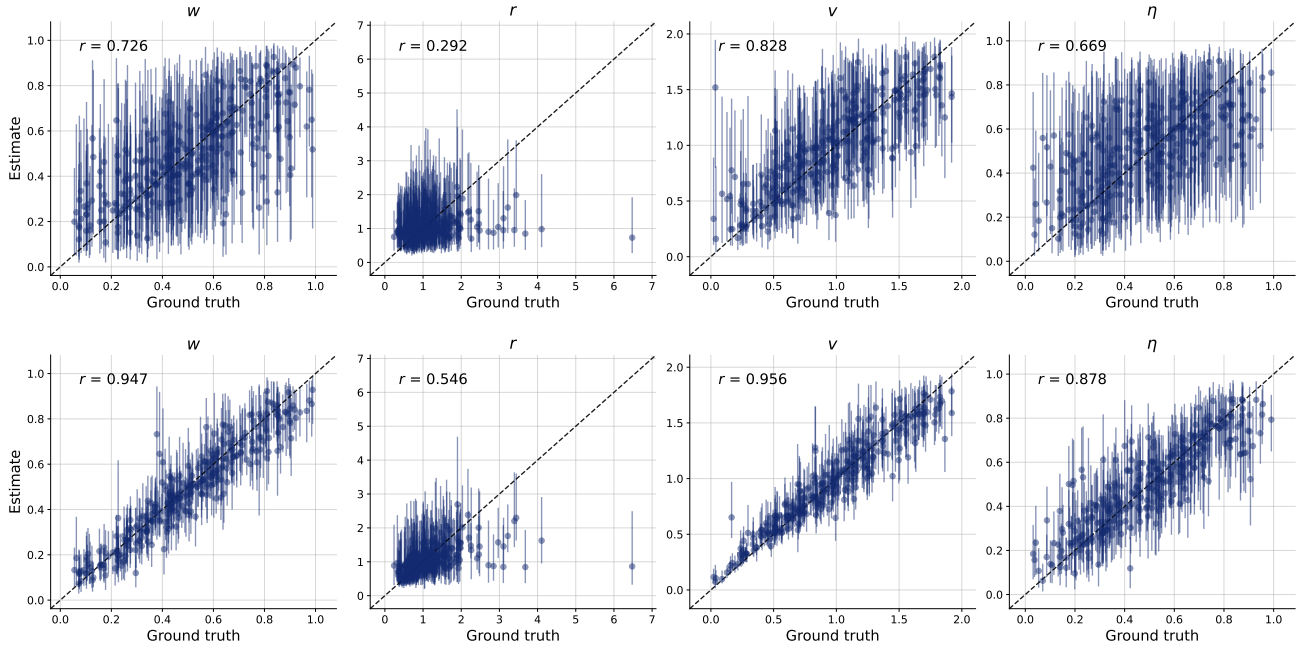


Figure 2: Parameter recovery for 3,000 (top) and 30,000 (bottom) offline training simulations. Points show posterior medians and bars marginal 95% credible intervals.

Table 1: Recovery and calibration for the revised model under two fixed offline training budgets, using a shared independent test bank. NRMSE uses prior-only prediction error as its normalizer; NRMSE and contraction are medians across test simulations.

	$N = 3{,}000$				$N = 30{,}000$			
Metric	w	r	v	η	w	r	v	η
Calibration error (ECE)	0.008	0.043	0.038	0.036	0.084	0.050	0.076	0.052
NRMSE	0.606	0.720	0.486	0.645	0.266	0.531	0.239	0.346
Posterior contraction	0.669	0.610	0.777	0.545	0.943	0.784	0.949	0.885

Figure 2 compares simulated parameters with posterior estimates. We report the median, across test simulations, of posterior-draw RMSE normalized by the median RMSE of prior-only predictions bootstrapped from the same test truths. Posterior contraction is the median clipped proportional reduction in variance relative to the empirical prior variance; both diagnostics use the same saved posterior draws as the plots. From 3,000 to 30,000 training simulations, NRMSE changes as follows: w: 0.606 to 0.266, r: 0.720 to 0.531, v: 0.486 to 0.239, η: 0.645 to 0.346.

Posterior contraction changes as follows: w: 0.669 to 0.943, r: 0.610 to 0.784, v: 0.777 to 0.949, η: 0.545 to 0.885. Calibration and contraction describe different properties: small ECE can coexist with broad, weakly informative posteriors. These results assess recovery under the specified simulated model and observation design; they do not establish structural non-identifiability of sensing radius. Differences from the original main-paper estimates also reflect revised dynamics, observables, architecture, and metric definitions.